EZEQUIEL DE BRAGA SANTOS

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ezequielebs

Research Interests

Biostatistics, Bayesian and computational statistics, power priors, epidemiological forecasting.

Academic Background

› M.Sc. in Applied Mathematics and Data Science – Academic Performance: A 2025 – Present
Fundação Getulio Vargas (FGV/EMAp)
Rio de Janeiro, Brazil

Expected graduation: March 2026.

› B.Sc. in Applied Mathematics – Academic Performance: 9.08/10 2021 – 2024
Fundação Getulio Vargas (FGV/EMAp)
Rio de Janeiro, Brazil

Research Experience & Leadership

› Project Lead, Disease Forecasting Pipeline 03/2025 – Present
FGV EMAp / Mosqlimate Project

Led development of models to produce dengue and chikungunya forecasts for the 3rd Infodengue-Mosqlimate Dengue Challenge.

› Team Lead, World Cup 2026 Probabilistic Forecasting Project 2025 – Present
FGV EMAp

Led a team building Bayesian models to predict World Cup 2026 scores.

› Research Assistant 03/2025 – Present
FGV EMAp

Modeling dengue cases for the Mosqlimate project. Producing tutorial for the hd-bayes R package.

Teaching Experience

› Teaching Assistant 02/2023 – 07/2026
FGV EMAp

Numerical Linear Algebra, Statistical Inference, and Statistical Modeling.

Activities

› Presentations

2025 **Principled and Modern Bayesian Analysis with the Normalized Power Prior**
Poster presentation at the National Congress of Applied and Computational Mathematics (CNMAC) and at Statistics Week at the Federal Fluminense University (SEMEST).

› Participation on Congresses

2025 **Research in Options 2025 (RiO)**

2025 **E-Vigilância 2025**

2025 **II International Colloquium on Mathematical Modelling in Epidemiology**

› Scientific Initiation Programs

09/2023 – 08/2024 **Institutional Scientific Initiation Scholarship Program (PIBIC)**
FGV EMAp
Modeling and ranking of clubs and players in the Brazilian football championship.

09/2021 – 08/2023 **Scientific Initiation and Master's Program (PICME)**
FGV
Completed graduate-level coursework at FGV EMAp.

› Working Papers

In preparation **From priors to performance: enhancing statistical efficiency with Bayesian dynamic borrowing**
joint work with Xinxin Chen, Joseph G. Ibrahim, Luiz Max Carvalho and Ethan M. Alt

In preparation **Normalised power prior for multiple historical datasets**
joint work with Yure Oliveira Santos, Joseph G. Ibrahim and Luiz Max Carvalho

In preparation **Variational approximations for the normalised power prior**
joint work with Yure Oliveira Santos, Xinxin Chen, Diego Mesquita, Joseph G. Ibrahim and Luiz Max Carvalho

In preparation **A predictive approach to the power prior**
joint work with Yure Oliveira Santos, Joseph G. Ibrahim and Luiz Max Carvalho

In preparation **Evaluating the drivers of Leishmaniasis in São Paulo, Brazil: a modelling study**
joint work with Iara Castro, Isaque Vieira Machado Pim, Cleber V. Santos, Claudio J. Struchiner and Luiz Max Carvalho

In preparation **Player-level skill models for Brazilian football**
joint work with Igor Michels, Moacyr Silva and Luiz Max Carvalho

Skills

› **Programming:** Python, R, MATLAB, Scilab, Julia

› **Statistical & Computational:** Bayesian and computational statistics, statistical modeling and inference (football & epidemiology), machine learning, numerical methods, linear algebra, stochastic processes and calculus

› **Languages:** Portuguese (native), English (intermediate–advanced)

Awards & Recognition

🏆 Scholarship Recipient - FGV EMAp (Center for Development of Mathematics and Sciences), 2021 – 2024.

🏆 Bronze Medalist, Brazilian Public School Mathematics Olympiad (OBMEP) - Invited to entrance exam for outstanding performance in OBMEP, 2019.

🏆 Silver Medalist, Brazilian Public School Mathematics Olympiad (OBMEP) - 2017, 2018.